

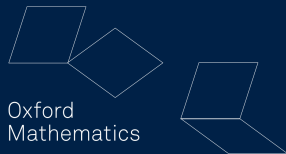
Differentiable Electrochemical Simulation for Gradient-Based Parameter Inference



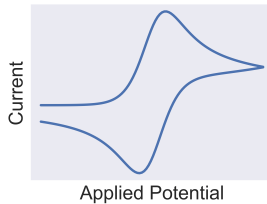
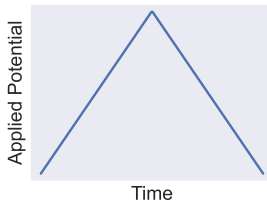
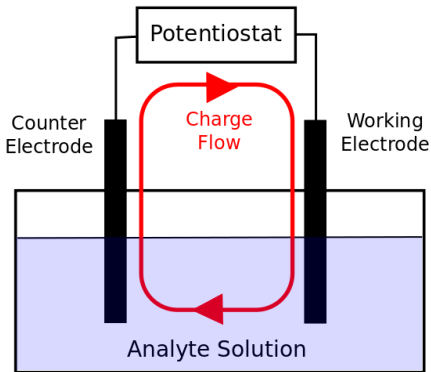
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OMMS Dissertation Presentation



Electrochemistry and Cyclic Voltammetry



Mathematical Description

► Diffusion:

$$\text{► } \frac{\partial c_A}{\partial t} = D_A \frac{\partial^2 c_A}{\partial x^2}$$

$$\text{► } \frac{\partial c_B}{\partial t} = D_A \frac{\partial^2 c_B}{\partial x^2}$$

► Initial Condition: $c_A(0, x) = c_A^*$, $c_B(0, x) = c_B^*$

► Infinite Boundary Conditions:

$$\text{► } c_A(t, x) \rightarrow c_A^* \text{ as } x \rightarrow \infty$$

$$\text{► } c_B(t, x) \rightarrow c_B^* \text{ as } x \rightarrow \infty$$

► Electrode Boundary Condition (Butler-Volmer):

$$\text{► } D_A \frac{\partial c_A}{\partial x} \bigg|_{x=0} = k_0 \left[c_A(t, 0) e^{-\alpha \frac{F}{RT} (E(t) - E_f^0)} - c_B(t, 0) e^{(1-\alpha) \frac{F}{RT} (E(t) - E_f^0)} \right]$$

$$\text{► } D_B \frac{\partial c_B}{\partial x} \bigg|_{x=0} = -D_A \frac{\partial c_A}{\partial x} \bigg|_{x=0}$$

► Current: $I = FAj_{0,A}$ where $j_{0,A} = -D \frac{\partial c_A}{\partial x} \bigg|_{x=0}$

Dimensionless Mathematical Description

Assuming $D_A = D_B$

► **Diffusion:** $\frac{\partial C_A}{\partial t} = \frac{\partial^2 C_A}{\partial x^2}$

► **Initial Condition:** $C_A(0, X) = 1$

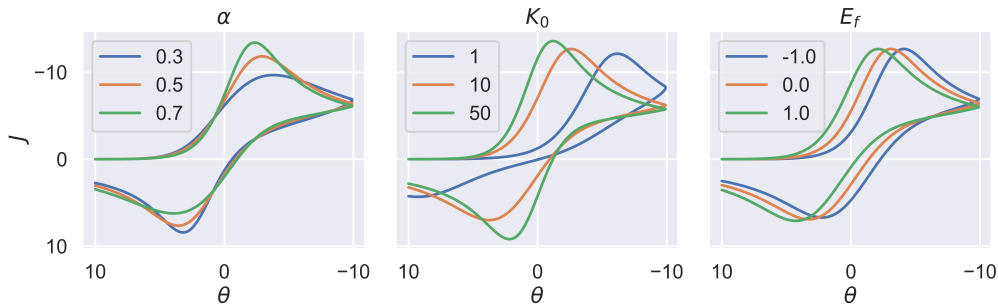
► **Semi-Infinite Boundary Conditions:** $C_A(t, X_{\max}) = 1$

► **Electrode Boundary Condition (Butler-Volmer):**

$$\left. \frac{\partial C_A}{\partial X} \right|_{X=0} = K_0 \left[C_A(T, 0) e^{-\alpha(\theta(T) - \theta_f^0)} - (1 - C_A(T, 0)) e^{(1-\alpha)(\theta(T) - \theta_f^0)} \right]$$

► **Current:** $I = \pi \epsilon F D_A c_A^* J_{0,A}$ where $J_{0,A} = -D \left. \frac{\partial C_A}{\partial X} \right|_{X=0}$

Parameter Effects



Forward and Inverse Problem

- ▶ Let $\phi = (\alpha, K_0, \theta_f) \in \Phi \subseteq \mathbb{R}^3$ be the parameter vector
- ▶ The finite difference solver defines a map:

$$\mathcal{P} : \Phi \rightarrow \mathcal{F}, \quad \mathcal{P}(\phi) = J(T)$$

where $\mathcal{F}$ is the space of dimensionless flux trajectories

- ▶ **Forward Problem:** Given ϕ , compute $\mathcal{P}(\phi) = J(T)$
- ▶ **Inverse Problem:** Given noisy observations $y \approx J(T)$, recover

$$\phi^* = \arg \min_{\phi \in \Phi} \|\mathcal{P}(\phi) - y\|$$

- ▶ The inverse problem is generally **ill-posed**

Bayesian Inference

- ▶ **Noise model:** $y_i = \mathcal{P}(\phi)_i + \varepsilon_i$, $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$
- ▶ **Prior:** Uninformative uniform prior over Φ , so $P(\phi) \propto 1$
- ▶ **Posterior:** $P(\phi | y) \propto L(y | \phi) \cdot P(\phi) \propto L(y | \phi)$
- ▶ **Likelihood:**

$$L(y | \phi) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathcal{P}(\phi)_i)^2}{2\sigma^2}\right)$$

- ▶ **Log-likelihood:**

$$\begin{aligned} \ell(\phi | y) &= -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \mathcal{P}(\phi)_i)^2 \\ &\propto -\frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \mathcal{P}(\phi)_i)^2 \end{aligned}$$

- ▶ **Goal:** Approximate the posterior $P(\phi \mid y)$ using samples $\{\phi^{(t)}\}_{t=1}^T$ from a Markov chain
- ▶ **Convergence:** Under mild regularity conditions, the empirical distribution of the chain converges to $P(\phi \mid y)$ as $T \rightarrow \infty$
- ▶ **Proposal:** At each step, candidate parameters ϕ_{cand} are drawn from a proposal distribution $q(\phi_{\text{cand}} \mid \phi^{(t)})$
- ▶ **Metropolis-Hastings acceptance:** Accept ϕ_{cand} with probability

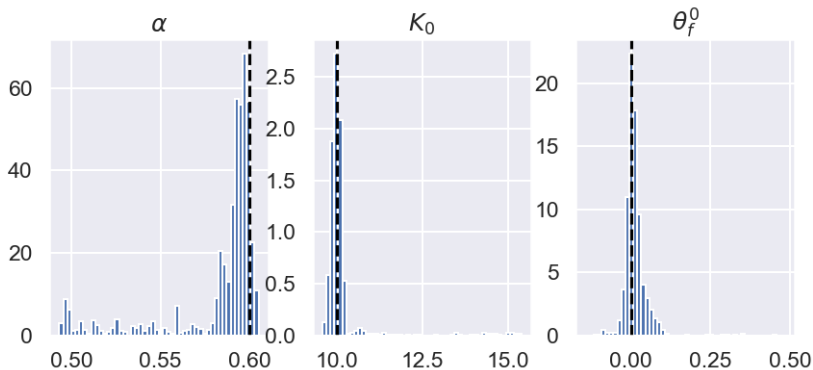
$$r = \min \left\{ \frac{P(\phi_{\text{cand}} \mid y)}{P(\phi^{(t)} \mid y)}, 1 \right\}$$

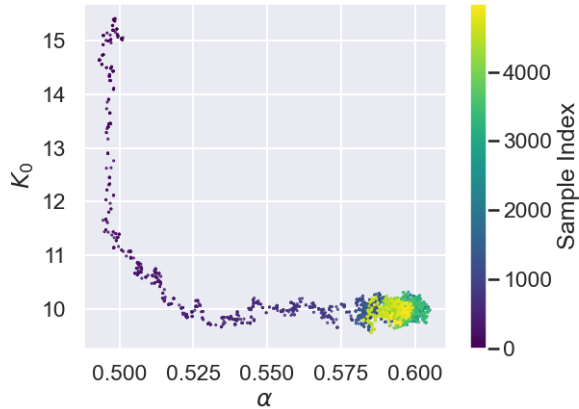
otherwise set $\phi^{(t+1)} = \phi^{(t)}$

Random-Walk Metropolis-Hastings

► Proposal:

$$q(\phi_{\text{cand}} \mid \phi^{(t)}) = \mathcal{N}(\phi^{(t)}, \Sigma)$$

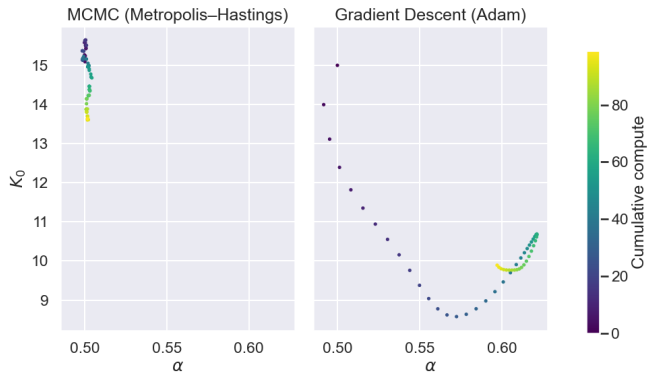




Differentiable Simulator

- ▶ **Why $\mathcal{P}$ is differentiable:** The finite difference discretisation reduces the PDE to a sequence of linear solves and pointwise nonlinear operations (Butler-Volmer), all of which are differentiable
- ▶ **Gradient of the log-likelihood:** We require $\nabla_{\phi} \ell(\phi | y)$
- ▶ **Reverse mode AD:** Since ℓ is scalar, reverse mode AD computes $\nabla_{\phi} \ell$ in a single backward pass through $\mathcal{P}$, at comparable cost to a forward evaluation
- ▶ **Applications:**
 - ▶ **Burn-in removal:** Gradient descent on $-\ell(\phi | y)$ to initialise the chain near the posterior mode
 - ▶ **Hamiltonian Monte Carlo:** Gradient informed MCMC yielding faster mixing and better exploration of $P(\phi | y)$

Burn-in removal

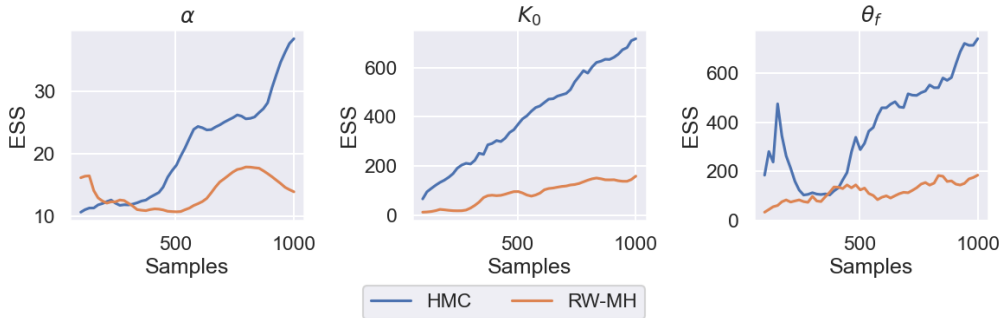


Hamiltonian Monte Carlo

- ▶ Augments parameter space with auxiliary momentum variables, using $\nabla_{\phi}\ell$ to make informed moves through the posterior
- ▶ **Effective Sample Size:** Autocorrelated samples carry less information than independent draws — ESS quantifies the equivalent number of independent samples:

$$\text{ESS} = \frac{T}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

where ρ_k is the autocorrelation at lag k



Further Results

- ▶ **Higher dimensional inference:** Sampling and numerical results for experiments with a greater number of parameters to infer
- ▶ **Non-linear PDEs:** Sampling and numerical results for electrochemical models requiring Newton's method or linearisation at each time step of the finite difference solver